Spherical Astronomy, Coordinates, and Time

yo where is bro at

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Prerequisite Knowledge

- Basic trigonometric identities (angle sum/difference formulas)
- Basic vectors: dot and cross products, and their geometric meaning
- Basic differentiation and integration (chain rule, related rates)
- Braincells

1 Introduction

Imagine two astronomers, one in Manila and one in Manchester, both looking at the same star at the same instant. If each of them describes the star's position as "high up and a bit to the left," they are talking about completely different directions in space — their local "up" and "left" depend on where they happen to be standing on a rotating, spherical Earth. So the very first problem in celestial navigation is not really about stars at all: it's about agreeing on a shared language for direction. That's what a coordinate system is — a rulebook, so that two people anywhere on Earth, or across time, can point at the same thing in the sky and describe it the same way.

2 Definition of Terms

- **Celestial sphere** — an imaginary sphere, centered on the Earth, with an infinite radius. This is a genuinely useful trick: since every star is so absurdly far away compared to the differences in their actual distances, we can pretend they're all painted on the inside of one giant dome and worry only about *direction*, never distance.
- **Zenith (Z)** — the point on the celestial sphere directly above an observer.
- **Nadir** — the point directly below the observer, 180 from the zenith.
- **Latitude** — measured from 0 (equator) to 90 (either pole). The set of points sharing a latitude is a **parallel of latitude**; the best-known ones are the tropic of cancer ($23.5N$), the equator (0), the arctic circle ($66.5N$), the antarctic circle ($66.5S$), and the tropic of capricorn ($23.5S$).
- **Longitude** — measured 0 to 180 east or west of the prime (Greenwich) meridian.
- **Great circle** — a circle drawn on a sphere's surface that shares the sphere's own center, and therefore has the sphere's own radius. It's the "straightest possible path" you can walk while confined to the surface of a sphere.
- **Local meridian** — the great circle on the observer's sky running from the north cardinal direction, through the zenith, to the south cardinal direction. Every longitude on Earth defines its own local meridian.

3 Horizontal and Equatorial Coordinates

3.1 Horizontal Coordinate System (a.k.a Alt-Az)

The most natural coordinate system is also the most personal one: describe a star by how high it is above your own horizon, and in which compass direction. That's the **horizontal (Alt-Az)** system.

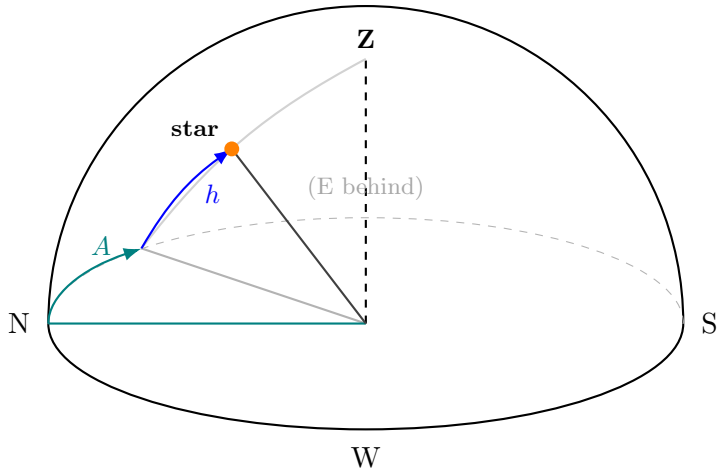


*Figure 1: The observer's sky bubble. The **altitude** h is measured from the horizon up to the zenith Z (0–90); the **azimuth** A is measured along the horizon from north, going east (0–360).*

The catch: horizontal coordinates are tied entirely to *you*. The instant you change location, or the instant time passes and the Earth turns beneath the star, both h and A change. Two observers looking at the same star simultaneously will report completely different Alt-Az values. That makes this system perfect for pointing a telescope, and useless for cataloguing stars.

One important thing you probably wanna remember: an observer standing exactly at a pole has the celestial equator running right along their horizon; an observer standing on the equator sees the north celestial pole sitting exactly on their north horizon, and the south celestial pole on their south horizon.

3.2 Equatorial Coordinate System (a.k.a RA and Dec)

To catalogue a star once and have that description remain valid for every observer everywhere, we need a coordinate system that doesn't care where you're standing — one tied to the sky itself. That's the **equatorial** system.

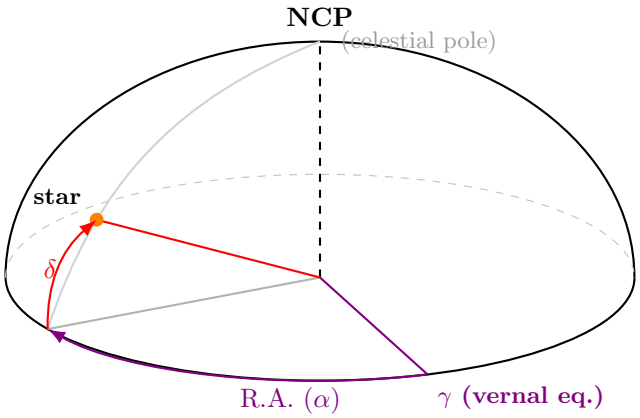


*Figure 2: The equatorial coordinate system, viewed from "above" the pole. The **declination** δ is the star's angular distance from the celestial equator (0–90, like a celestial latitude). The **right ascension** (R.A., α) is measured eastward from a fixed reference point on the celestial equator, the **vernal equinox** (γ) — like a celestial longitude, except it never rotates with the observer.*

Declination is essentially a star's "co-latitude on the sky," and because stars are so overwhelmingly distant, we can treat their declinations as fixed for decades at a time. The one glaring exception is our own Sun: because it's so much closer, and because Earth's axis is tilted by 23.5 relative to its orbit, the Sun's declination sweeps from $23.5S$ to $23.5N$ and back over the course of a year. That sweep is the entire reason we have seasons.

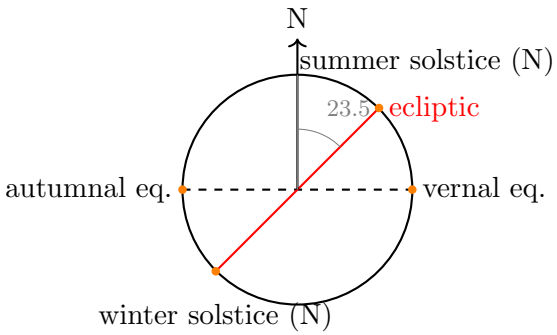


Figure 3: The Sun's declination over the year, shown as the tilt of the ecliptic (the Sun's apparent path) against the celestial equator. At the summer solstice the Sun sits at its northernmost declination ($+23.5$, over the tropic of cancer); at the winter solstice its southernmost (-23.5, over the tropic of capricorn); the convention flips for the southern hemisphere. At either equinox the Sun crosses the celestial equator, $\delta = 0$.

Right ascension plays the role of celestial longitude, except it's anchored to the vernal equinox rather than to Greenwich, and it's conventionally expressed in *hours* rather than degrees: since a full circle is 360 and there are 24 hours in that circle, every 15 of R.A. equals 1 hour. An R.A. of 20, for instance, is 1^h20^m.

3.3 Synthesis of the Horizontal and Equatorial Coordinate Systems

To summarize: horizontal coordinates (h, A) are defined with respect to the *observer*, change constantly, and are what you'd actually dial into a telescope mount. Equatorial coordinates (δ, α) are defined with respect to the *sky*, stay fixed for a star over human timescales, and are what you'd find printed in a star atlas. Neither system is "better" — they simply answer different questions, and the whole next stretch of this handout is about building the bridge between them.

4 Hour Angle (H)

Before we can build that bridge, we need one more ingredient: a way of measuring *how far a star has traveled across the sky since it last crossed your local meridian.* That's the **hour angle**, H — the angular difference between two meridians (the star's own meridian and your local meridian), measured to the west of your local meridian.[1]

[1] Some resources measure hour angle to the east instead; either convention works, as long as you're consistent.

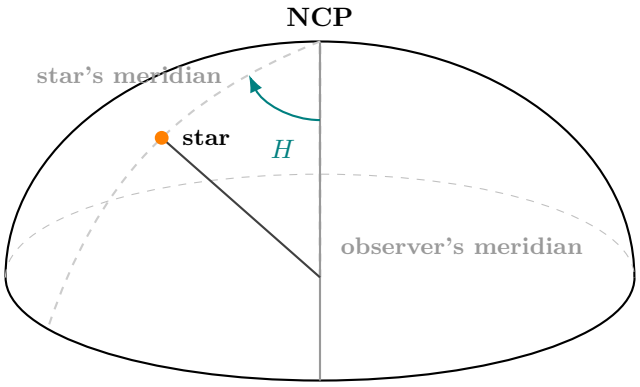


Figure 4: Hour angle H. The zero-value of H is when the star sits exactly on the observer's local meridian; H then grows as the star drifts west, which is the natural direction to treat as "forward in time," since stars rise in the east and set in the west.

5 Sidereal Time

The Earth spins on its axis once every $23^h56^m4^s$ — a **sidereal day** — as measured against the fixed background stars (a few minutes shorter than the familiar 24-hour solar day, because the Earth also has to "catch up" a little extra each day to account for its own orbital motion around the Sun). Over one sidereal day, every right ascension value on the celestial sphere sweeps past your local meridian exactly once. The R.A. value currently sitting on your local meridian is called your **local sidereal time (LST)**.

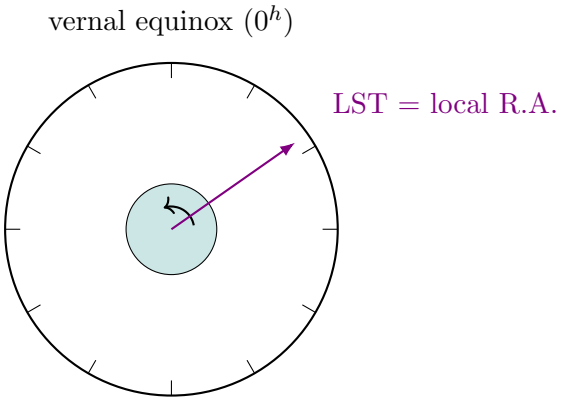


Figure 5: The R.A. "clock face" stays fixed against the stars while the Earth (center) turns underneath it. The arrow marks whichever R.A. value currently coincides with the observer's local meridian — that value is the LST.

Since the local meridian's hour angle is by definition zero for whatever is currently sitting on it, and since the star's own hour angle plus its right ascension must together account for the meridian's total sweep, we get a simple and extremely useful relation:

$$\boxed{LST = H + \alpha.}$$

Knowing the right ascension of any star you can see, plus its current hour angle, immediately hands you your own local sidereal time.

6 Circumpolarity

6.1 Upper and Lower Culminations

A star is in **upper culmination** when it transits the local meridian at its *maximum* altitude for the day; it's in **lower culmination** when it transits at its *minimum* altitude. A star that rises and sets only ever shows an upper culmination. A star that never dips below the horizon crosses the local meridian twice a day (once high, once low) and is called **circumpolar**. (We'll fix the sign convention that the northern hemisphere is positive.)

6.2 Maximum and Minimum Altitudes

At upper culmination, the star, the NCP, and the zenith all lie on the same meridian, so the maximum altitude is simply 90 minus the absolute difference between declination and latitude:

$$\boxed{h_u = 90 - |\phi - \delta|.}$$

At lower culmination, a symmetric argument (the star now sitting on the *far* side of the pole from the zenith) gives the minimum altitude of a circumpolar object:

$$\boxed{h_l = |\phi + \delta| - 90.}$$

These two equations are secretly a linear system in ϕ and δ. Subtracting them cancels the declination and leaves an equation purely in latitude:

$$h_u - h_l = 180 - 2\phi \quad \Longrightarrow \quad \boxed{\phi = \frac{h_l - h_u + 180}{2}.}$$

Adding them instead cancels the latitude:

$$\boxed{\delta = \frac{h_u + h_l}{2}.}$$

(If either solved angle exceeds 90 — say, 100 south — take its difference from 90 and subtract that from 90 again: 100 south becomes 80 south.)

6.3 Declination of Non-Rising and Non-Setting Stars

A star never sets — is circumpolar — at latitude ϕ exactly when its minimum altitude is positive, $h_l > 0$:

$$\boxed{|\phi + \delta| > 90.}$$

Symmetrically, a star never rises at all when its *maximum* altitude is negative, $h_u < 0$:

$$\boxed{|\phi - \delta| > 90.}$$

7 Sun Rising and Set Times

We can solve for the local time of sunrise or sunset with a single compact formula:

$$\boxed{\cos H = -\tan\phi \tan\delta,}$$

where δ is the Sun's declination and ϕ the observer's latitude (we'll derive this properly once we have spherical trigonometry in hand, in Section 12.1.3 — for now, take it on faith). Converting the solved H to hours (every 360 is 24^h) and treating sunrise and sunset as roughly symmetric about local noon (ignoring the Sun's slow declination drift over the course of a single day) gives the rise and set times directly.

8 The Horizon at Heights

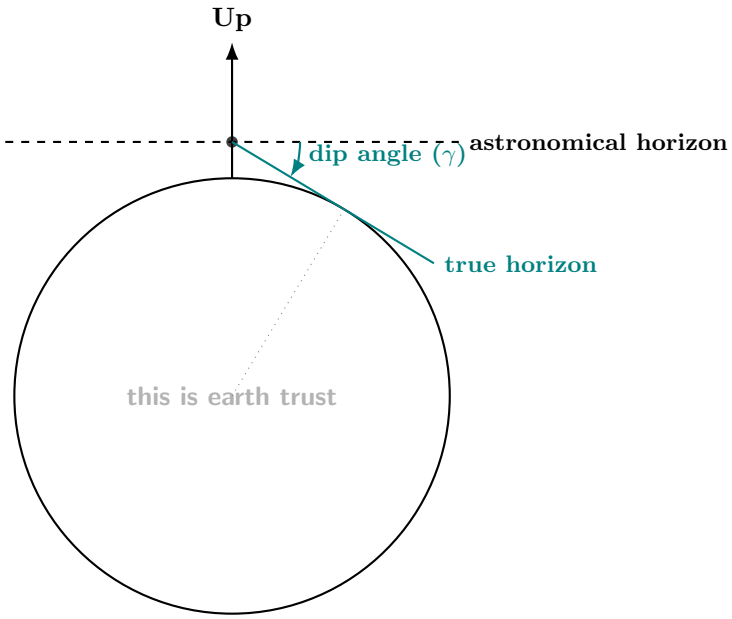


Figure 8: The dip angle. From height h above sea level (the gray dot, raised above the Earth's surface), the observer's true horizon — the line of sight that just grazes the Earth's curve — dips below the eye-level astronomical horizon.

At the top of a mountain, you can see further below eye-level than you could at sea level. The **dip angle**, the extra angle of sight gained from height h, is

$$\boxed{\Theta_{dip} = \cos^{-1}\left(\frac{R}{R+h}\right),}$$

where R is Earth's radius — a plain right-triangle relation between the observer's height above the surface and the tangent line to that surface.

9 The Ecliptic

The **ecliptic** is the great circle traced by the Sun's apparent path across the celestial sphere over one year (equivalently, it's the plane of Earth's orbit, projected onto the sky). As it sweeps around, it passes through a set of constellations — the **zodiac constellations**.

Zodiac Constellation	**Dates**
Capricornus	Jan 20 – Feb 16
Aquarius	Feb 16 – Mar 11
Pisces	Mar 11 – Apr 18
Aries	Apr 18 – May 13
Taurus	May 13 – June 21
Gemini	June 21 – July 20
Cancer	July 20 – Aug 10
Leo	Aug 10 – Sep 16
Virgo	Sep 16 – Oct 30
Libra	Oct 30 – Nov 23
Scorpius	Nov 23 – Nov 29
Ophiuchus	Nov 29 – Dec 17
Sagittarius	Dec 17 – Jan 20

The Sun genuinely spends at least 20 days each year passing through Ophiuchus — more time than it spends in some of the "official" twelve — but because the ecliptic only barely clips the constellation's corner on a sky map, Ophiuchus is conventionally left off the list.

10 Spherical Triangle

10.1 Definition of a Spherical Triangle

Everything so far has been definitions and one-off formulas. To actually *derive* the relationships between horizontal and equatorial coordinates, we need a general-purpose tool: the **spherical triangle**.

Take three great circles on a sphere. Generically, each pair of them crosses at two (antipodal) points, and the arcs connecting three well-chosen crossing points bound a triangle drawn entirely on the sphere's curved surface. That's a spherical triangle — and it is emphatically *not* the flat triangle from your geometry class. Its three "sides" are arcs of great circles, curving along the sphere, and its interior angles (the angles between the *planes* of the great circles meeting at each vertex) don't have to add up to 180 at all — in fact, as we'll see in Section 11.4, they always add up to *more*.

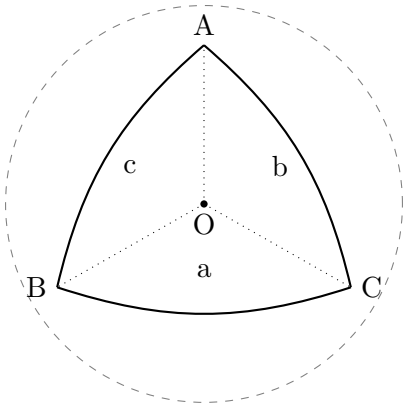


Figure 9: A spherical triangle. The dashed circle is the sphere itself; the solid curved arcs a, b, c are arcs of great circles on that sphere's surface, and O is the sphere's center. Lower-case letters name the arc lengths (equivalently, the central angles: the arc $a = \angle BOC$), and upper-case letters name the vertex angles — $\angle A$ is the angle between the planes AOB and AOC.

10.2 Visualizing the Relationship between Eqr-Hor Coordinate Systems using ST

Now let's put this shape to work. Let the sphere be the celestial sphere, and let its three vertices be the observer's zenith Z, the north celestial pole N, and a celestial object X.

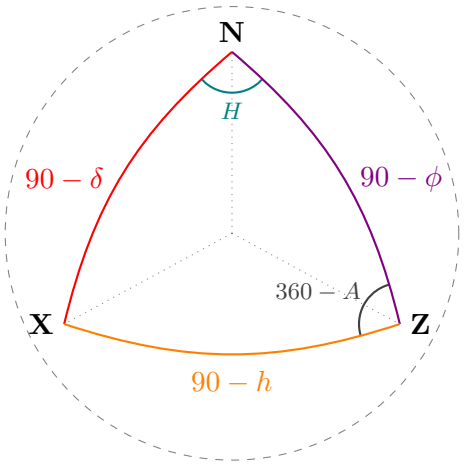


Figure 10: The spherical triangle NXZ that connects the north celestial pole, a celestial object, and the observer's zenith. This one triangle is the entire bridge between horizontal and equatorial coordinates.

Each side of this triangle is something we already recognize. The arc $\widehat{ZX}$ is the angular separation between the zenith and the object — the complement of altitude, $\widehat{ZX} = 90 - h$. The arc $\widehat{ZN}$ is the angular separation between the zenith and the pole, which is $90 - \phi$. And $\widehat{XN}$ is the complement of declination, $90 - \delta$.

What about the vertex angles? At the pole N, the angle between the object's meridian and the observer's meridian is, by definition, the hour angle: $\angle N = H$. At the zenith Z, the angle works out to 360 minus the azimuth, because azimuth is conventionally measured increasing *eastward* while the angle in our triangle is measured the other way around:

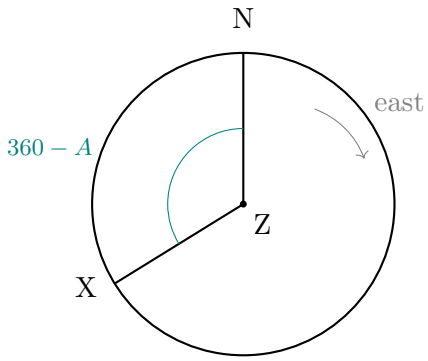


Figure 11: A flat top-down view of the observer's horizon, used only to pin down the azimuth convention. Since azimuth increases eastward while our triangle's vertex angle at Z opens the other way, $\angle Z = 360 - A$.

We now have every side and every angle of the NXZ triangle expressed in terms of quantities we already understand. All that's left is to plug this triangle into the machinery of spherical trigonometry — which is exactly what Section 12 does.

11 Spherical Trigonometry

11.1 Law of Cosines

Proposition: for a spherical triangle with sides a, b, c and opposite angles A, B, C,

$$\boxed{\cos a = \cos b \cos c + \sin b \sin c \cos A.}$$

Sketch of proof: place the sphere's center at the origin of ordinary 3D space, with radius 1, and put vertex A at the north pole $(0, 0, 1)$. Vertices B and C then sit somewhere else on the unit sphere, and we can write their Cartesian coordinates using ordinary spherical coordinates (an angle around the pole, and an angle down from it). The straight-line (chord) distance between B and C can be computed two different ways: once from the ordinary flat-space distance formula in terms of their coordinates, and once from the flat-triangle law of cosines applied to the isoceles triangle OBC (both sides length 1, included angle a). Setting these two expressions for the same chord equal to each other, and grinding through the algebra (it eventually produces a term $\cos(\phi_B - \phi_C)$, which the angle-difference identity $\cos(x-y) = \cos x \cos y + \sin x \sin y$ unpacks), lands you exactly on the boxed formula above. The full algebra is mechanical rather than illuminating, so we won't reproduce every line here — the point worth keeping is that the spherical law of cosines is, underneath, ordinary 3D Pythagorean-style geometry, just expressed in angles instead of lengths.

11.2 Law of Sines

Proposition:

$$\boxed{\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c}.}$$

The proof requires more vector machinery than we need here, but the formula itself is straightforward to use: any spherical triangle where you know two angles and a side (or two sides and an angle) opposite each other can be solved with this single line.

11.3 Napier's Circle

A **right spherical triangle** is one with a 90 vertex. This special case unlocks a set of shortcut formulas, and there's a slick mnemonic — **Napier's circle** — for remembering all of them at once.

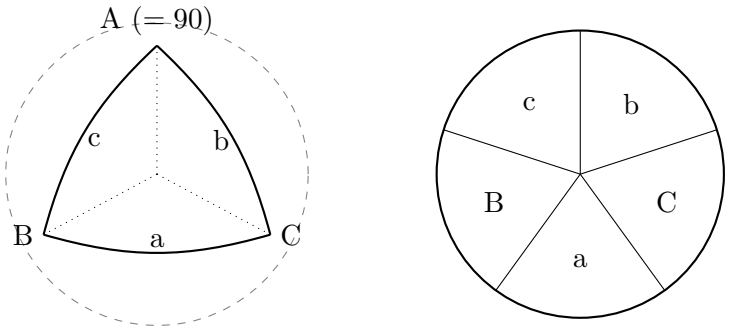


Figure 12: Left — a right spherical triangle, with the curved sides drawn exactly as in Figure 9, just with one vertex (A) fixed at 90*. Right — Napier's circle: the five* other *angular elements of the triangle (everything except the known* 90 *angle), arranged around a wheel in the same order they appear going around the triangle.*

Two rules fall out of this wheel:

Tan-ad rule: the sine of any element equals the product of the tangents of its two *adjacent* (neighboring) elements on the wheel:

$$\boxed{\sin B = \tan c \tan a.}$$

Co-op rule: the sine of any element equals the product of the cosines of its two *opposite* (non-adjacent) elements on the wheel:

$$\boxed{\sin B = \cos b \cos C.}$$

(B can be swapped for any of the five elements, as long as you correctly identify its adjacent or opposite neighbors on the wheel.) You don't strictly need to memorize either formula — just know how to draw the wheel, and read the relationship straight off it.

11.4 Area of a Spherical Triangle

We already hinted that a spherical triangle's angles sum to more than 180. The excess, $E = 180 - A - B - C$, isn't just a curiosity — it's directly proportional to the triangle's surface area:

$$\boxed{A_{area} = ER^2.}$$

This makes intuitive sense at the extremes: if a spherical triangle's angles summed to exactly 180, this formula says its area would be exactly zero. So any spherical triangle with actual, nonzero area is guaranteed to have some genuine angular excess above 180 — which is really just another way of saying the triangle's sides must be visibly curved, not flat.

12 Applications of Spherical Trigonometry in Astronomy

12.1 Relationships between Eqr and Hor Coordinate System

Time to cash in the NXZ triangle from Section 10.2.

12.1.1 Solving for the Altitude from Equatorial Coordinates

Apply the spherical law of cosines to arc $\widehat{ZX}$ (opposite vertex N, included angle H, adjacent sides $\widehat{ZN} = 90 - \phi$ and $\widehat{XN} = 90 - \delta$):

$$\cos(90 - h) = \cos(90 - \delta)\cos(90 - \phi) + \sin(90 - \delta)\sin(90 - \phi)\cos H.$$

Using $\cos(90 - x) = \sin x$ and $\sin(90 - x) = \cos x$ throughout,

$$\boxed{\sin h = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H.}$$

One single equation, and it connects all four of the parameters we've been building toward: altitude, declination, latitude, and hour angle.

12.1.2 Solving for the Declination from Horizontal Coordinates

Running the same law of cosines on the side $\widehat{XN} = 90 - \delta$ instead (opposite vertex Z, included angle $360 - A$, recalling $\cos(360 - A) = \cos A$):

$$\boxed{\sin\delta = \sin h \sin\phi + \cos h \cos\phi \cos A.}$$

12.1.3 Hour Angle Formula (Sunrise/Sunset)

Setting $h = 0$ (the object exactly on the horizon) in the altitude formula gives us back the sunrise/sunset relation promised back in Section 7:

$$0 = \sin\delta \sin\phi + \cos\delta \cos\phi \cos H \quad \Longrightarrow \quad \boxed{\cos H_\odot = -\tan\delta_\odot \tan\phi.}$$

Converting H to hours via $t(\text{hr}) = \frac{H}{360} \cdot 24^h$, the time of sunset works out to $12{:}00 + \frac{H_\odot}{360} \cdot 24^h$, and sunrise to $12{:}00 - \frac{H_\odot}{360} \cdot 24^h$ (assuming a circular orbit and constant solar declination over the course of one day).

12.2 Local Sidereal Time

Notice that of our four horizontal/equatorial parameters (h, A, δ, α), right ascension α never appeared in Sections 12.1.1–12.1.3. We don't need those formulas to bring α into the picture at all — we already built the connector back in Section 5:

$$\boxed{LST_{observer} = H_{object} + \alpha_{object}.}$$

And if we can solve for the Sun's own hour angle and right ascension, we get an even more useful version:

$$\boxed{LST = H_\odot + \alpha_\odot = H_{object} + \alpha_{object}.}$$

12.3 Local Time

Local time (LT), also called **mean solar time**, is the clock time tied to the Sun's hour angle, under the simplifying assumption that the Sun moves at a perfectly constant rate across the sky all year (ignoring the real eccentricity of Earth's orbit — more on that in Section 14.4):

$$\boxed{LT = H_{\odot,\text{hours}} + 12^h.}$$

12.4 Extra Horizon Angle

We already met the dip angle in Section 8; here's the rigorous version, using x for height and $R_\oplus$ for Earth's radius:

$$\boxed{\cos\Theta = \frac{R_\oplus}{R_\oplus + x}.}$$

At a given height x, sunrise and sunset genuinely shift, because the Sun must clear this extra angle Θ before it reaches the *true* (elevated) horizon rather than the eye-level (astronomical) one. There's also **atmospheric refraction** — light bending as it passes through the atmosphere — which lifts the apparent position of objects near the horizon by about $34''$; we'll usually neglect it unless a problem says otherwise.

12.5 Sample Problems

Problem: Resolving Mizar and Alcor

A telescope's minimum angular resolution is $\Theta = 1.22\lambda/D$. Mizar ($\alpha = 13^h26^m16^s$, $\delta = 5451'33.2''$) and Alcor ($\alpha = 13^h24^m58.3^s$, $\delta = 5447'46.5''$) appear as a single point to the naked eye. What telescope diameter would be needed to resolve them, using $\lambda = 550\,nm$?

Converting both coordinates to decimal degrees and applying the spherical law of cosines to the angular separation Θ between the two stars,

$$\cos\Theta = \cos(90-\delta_A)\cos(90-\delta_M) + \sin(90-\delta_A)\sin(90-\delta_M)\cos(\alpha_A - \alpha_M)$$

gives $\Theta \approx 0.1968 = 3.4355 \times 10^{-3}$ rad. Plugging into $D = 1.22\lambda/\Theta$ gives $D_{min} \approx 1.95\,mm$ — comfortably smaller than a human pupil (2–8 mm), so most people can split these two stars with the naked eye if they focus.

Problem: Altitude of Eltanin from Quezon City

Eltanin sits at $\delta = 51.5N$. One night it transits the local meridian of Quezon City ($\phi = 14.5N$). Find its altitude.

At transit, $H = 0$, so the altitude formula collapses to $\sin h = \sin\delta\sin\phi + \cos\delta\cos\phi$, which is just $\cos(\phi - \delta)$. Plugging in gives $h = 53$ (and a zenithal distance of 37, its complement).

13 Spherical Trigonometric Relationships between Equatorial and Ecliptic Coordinates

We've related horizontal and equatorial coordinates; now let's add a third system: **ecliptic coordinates**, obtained by tilting the whole equatorial system by the axial tilt $\epsilon = 23.5$. A point's position is now given by **ecliptic latitude** β and **ecliptic longitude** λ (increasing eastward), and we call the pole of this system K, the ecliptic north pole, to distinguish it from the ordinary celestial pole N.

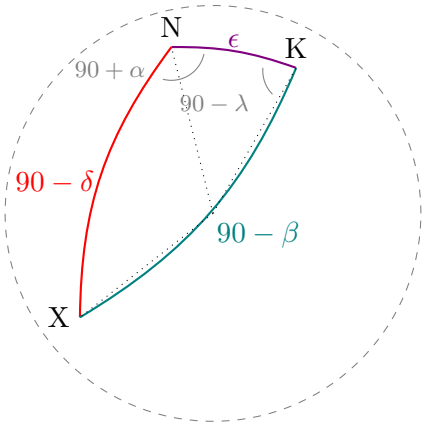


Figure 13: The spherical triangle connecting the celestial pole N, the ecliptic pole K, and an object X. The tilt angle ϵ sits right between the two poles.

13.1 Calculating the Ecliptic Latitude from the Ecl-Eqr Coordinates

Applying the law of cosines to this triangle:

$$\boxed{\sin\beta = \cos\epsilon\sin\delta - \cos\delta\sin\epsilon\sin\alpha.}$$

13.2 Calculating the Declination from Equatorial-Ecliptic Coordinates

Running the same triangle the other way:

$$\boxed{\sin\delta = \cos\epsilon\sin\beta + \cos\beta\sin\epsilon\sin\lambda.}$$

13.3 Emergent Formulas from the Ecliptic-Equatorial Coordinates

The Sun always sits exactly on the ecliptic plane, so its ecliptic latitude is always $\beta_\odot = 0$. Substituting that into the declination formula above immediately gives the sinusoidal law behind the seasons we sketched back in Figure 3:

$$\boxed{\sin\delta_\odot = \sin\epsilon\sin\lambda_\odot.}$$

Since the Sun's ecliptic longitude advances at an (approximately) constant rate through the year,

$$\boxed{\lambda = \frac{n}{365.242}\cdot 360,}$$

where n counts days since the vernal equinox and 365.242 is the length of the tropical year. One more emergent relation, whose derivation needs tools from Section 14.2:

$$\boxed{\tan\alpha_\odot = \cos\epsilon\tan\lambda_\odot.}$$

13.4 Sample Problem

Problem: The Galactic Pole and Vega

The northern galactic pole sits at equatorial coordinates $(\alpha_G, \delta_G) = (12^h51^m, 2708')$. Find its ecliptic latitude and longitude, then find the galactic longitude of Vega $(\alpha_v, \delta_v) = (18^h36^m56.3^s, 3847'1.3'')$, given the galactic center sits at $(17^h45^m40^s, -2900'28.1'')$.

Plugging (α_G, δ_G) into the two boxed formulas above gives $\beta_G = -08'$ and $\lambda_G = 30113'$. For Vega's galactic longitude, we build a second spherical triangle out of the galactic pole G, the galactic center C, and Vega V: the arc $\widehat{GC} = 90$ (the center sits on the galactic plane, by definition), $\widehat{GN} = 90 - |\delta_G|$, and $\widehat{CV}$ is found from the ordinary law of cosines applied to C and V's equatorial coordinates ($\widehat{CV} = 68.839$). One more application of the law of cosines, now to the triangle GVC, yields the galactic longitude $\iota_V = 67.53$.

14 Other Spherical Trigonometric Relationships

14.1 Four Parts Formula

One more relation, useful when you know an arc, its two adjacent pieces of information (an adjacent arc and adjacent angle), and want the opposite angle without going through two separate applications of the law of cosines:

$$\boxed{\cos c = \frac{\cos a \cos B + \sin B \cot C}{\sin a}.}$$

14.2 Rotational Coordinate Transformations

All of the formulas in Sections 12 and 13 can also be derived by a different route: rotating a 3D Cartesian coordinate system by the angle between two vertices of the triangle, and matching up coordinates before and after the rotation. Concretely, if A sits at the pole $(0, 0, 1)$ and B, C are two other vertices separated by arc α, then rotating the whole frame by α turns B's coordinates into C's, giving three simultaneous equations (one per axis) instead of just one. This is more work to set up than the law of cosines, but it pays off by handing you *three* related formulas at once instead of one. Applying this machinery to the NXZ triangle (letting $A = N$, $B = Z$, $C = X$) produces

$$\boxed{\cos H \cos \delta = \cos A \cos h \sin \phi + \sin h \cos \phi,} \qquad \boxed{\sin H \cos \delta = \sin A \cos h,}$$

alongside the altitude formula we already derived (Section 12.1.1) as the third component. Doing the same with the ecliptic triangle from Section 13 gives two further relations to sit alongside the ones we already found:

$$\boxed{\sin \lambda \cos \beta = \sin \delta \sin \epsilon + \cos \delta \cos \epsilon \sin \alpha,} \qquad \boxed{\cos \lambda \cos \beta = \cos \delta \cos \alpha.}$$

14.3 Proper Motion

Stars are not painted onto the celestial sphere forever — even ignoring Earth's own rotation, precession, and revolution, real stars slowly drift across the sky, typically taking decades to move even a single arcminute. This drift is called **proper motion**, and it splits naturally into two tangential components: motion along a declination circle, and motion along a meridian.

The meridian component is straightforward: $u_\delta = \Delta\delta/\Delta t$. The declination-circle component needs one extra piece of care, because a circle of constant declination δ has a smaller circumference than the celestial equator — shrunk by exactly a factor of $\cos\delta$ (the same effect that makes lines of longitude bunch up near Earth's poles). So

$$\boxed{u_\alpha = \frac{\Delta\alpha}{\Delta t}\cos\delta.}$$

Since these two components are tiny angles, the curvature between them is negligible over the relevant timescales, and we can add them with a plain Pythagorean theorem rather than the full spherical law of cosines:

$$\boxed{u = \sqrt{\left(\frac{\Delta\delta}{\Delta t}\right)^2 + \left(\frac{\Delta\alpha}{\Delta t}\cos\delta\right)^2}.}$$

14.4 Equation of Time

Mean solar time (Section 12.3) assumes the Sun crawls across the sky at a perfectly constant 15 per hour. Because Earth's orbit is actually elliptical, the Sun's real ("true") angular speed varies throughout the year, so the **true solar time** and the mean solar time slowly drift apart — by as much as 16 minutes. We call this drift the **equation of time**:

$$\boxed{E.T. = ST_{true} - ST_{mean}.}$$

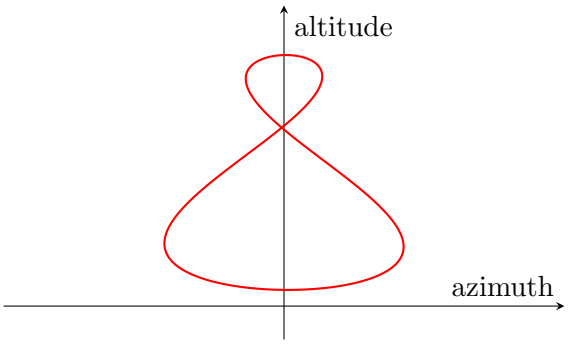


Figure 14: The analemma — the figure-eight traced on the sky by the Sun's position at the same clock time every day of the year, a direct visual record of the equation of time (its east-west width) combined with the seasonal swing in declination (its north-south length).